



TUNING GEOMETRY (§5; n=7 subjects). Second readout of the error field: does its direction coincide with the channels a voxel already encodes? Per voxel $\cos(E, u_i)$ (in-tuning if > 0) and the coupling of per-clip alignment to the reversal response vs a clip-shuffle null (FRAC of voxels $p < 0.05$; 0.05 = chance). Computed in the WHITENED basis (ridge-shrinkage mitigation) and, critically, re-run on the TOP-10% ncsnr voxels to rule out underpower. Top: candidate worlds, same plot (cos = purple, coupling frac-sig = green).

WHAT WE EXPECT (and why): If the violation is read out IN-TUNING: $\cos(E, u) > 0$ and coupling frac-sig $\gg 0.05$. If OFF-TUNING (a separate channel): $\cos \sim 0$ and coupling at chance. The high-SNR re-run is the discriminator -- if selecting reliable voxels does NOT raise the coupling above chance, the null is REAL, not underpowered.

READING (candidate; the call is yours): $\cos(E, u) \sim +0.002$ (≥ 0 chance) and coupling frac-significant ~ 0.049 vs the 0.05 chance floor across ROIs; high-SNR re-run frac-sig = 0.051. The error field is read out OFF-TUNING -- orthogonal to existing content tuning -- and selecting high-SNR voxels does NOT rescue it: the geometry null is REAL, not underpowered.